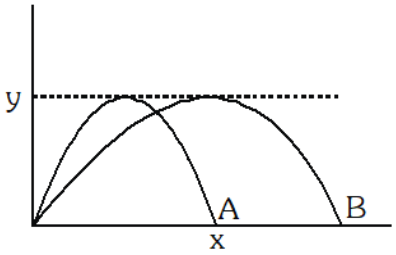
PART-1 : PHYSICS

SECTION-I (i)

1) Two particles are thrown from same point in same vertical plane as shown in figure



simultaneously. Then indicate the correct statement(s).

(A) Time of flight of B is less than that of A.

(B) Projection speed of B is greater than that of A.

(C) Horizontal component of velocity of B is greater than that of A.

(D) Vertical components of velocities of A and B are always equal through out the flight of A and B.

2) A ball is rolled off the edge of a horizontal table at a speed of 4 m/s. It hits the ground after 0.4 second. Which statement given below is true

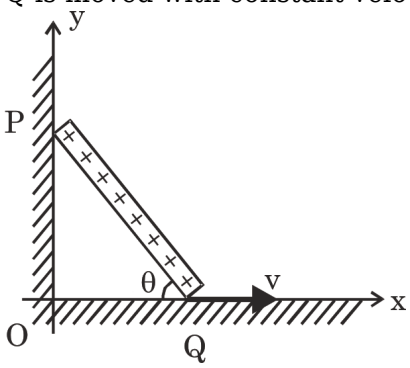
(A) The speed with which it hits the ground is 4.0 m/second

(B) It hits the ground at a horizontal distance 1.6 m from the edge of the table

(C) Height of the table is 0.8 m

(D) It hits the ground at an angle of 60° to the horizontal

3) A uniformely charged thin rod having uniform charge density is placed against a vertical wall as shown in figure. Its end Q is moved with constant velocity v and initially θ being 60° then before the



instant θ becomes zero.

(A) Magnitude of electric field at O first increases then decreases.

(B) Magnitude of electric field at O first decreases then increases.

(C) Magnitude of electric field at O remains constant.

(D) Direction of electric field at O continuously changes.

4) An oil drop has a charge -9.6×10^{-19} C and mass 1.6×10^{-15} gm. When allowed to fall, due to air resistance force it attains a constant velocity. Then if a uniform electric field is to be applied vertically to make the oil drop ascend up with the same constant speed, which of the following are correct? ($g = 10\ ms^{-2}$) (Assume that the magnitude of resistance force is same in both the cases)

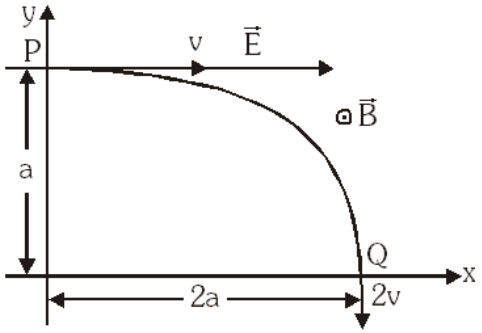
(A) The electric field is directed upward

(B) The electric field is directed downward

(C) The intensity of electric field is $\frac{1}{3} \times 10^2 NC^{-1}$

(D) The intensity of electric field is $\frac{1}{6} \times 10^5 NC^{-1}$

5) A particle of charge +q and mass m moving under the influence of a uniform electric field $E\hat{i}$ and uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$, Which of the following statement(s) is/are correct ?



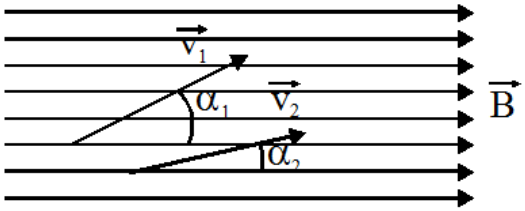
(A) $E = \frac{3}{4}\left[\frac{mv^2}{qa}\right]$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left[\frac{mv^3}{a}\right]$

(C) Rate of work done by the electric field at P is zero

(D) Rate of work done by both the fields at Q is zero

6) In a uniform magnetic field there are two charged particles 1 & 2 moving with velocities v_1 and v_2 respectively and carrying equal charges & equal mass with $|v_1| = |v_2| = v$. The velocity of one particle forms an angle α_1 with the direction of the field, while the other velocity forms an angle α_2, $(\alpha_1 > \alpha_2)$ Mark the CORRECT statement(s) :-



(A) Both the particles move along helical path

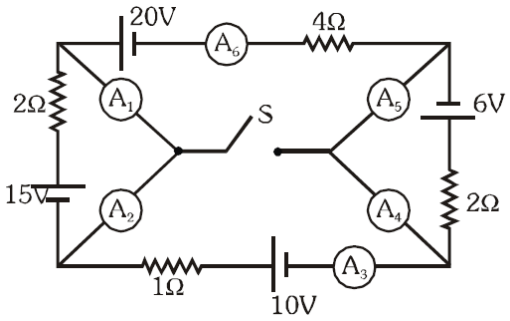
(B) Time period of circulation is same for both the particles

(C) Pitch of particle 1 is greater than that of particle 2

(D) Radius of helix is greater for particle 1

SECTION-I (ii)

Common Content for Question No. 1 to 2



In circuit shown all ammeters are ideal

1) If switch S is open, the ammeter (s) that read more than 10 A

(A) A_1 & A_2

(B) A_3 & A_4

(C) A_5 & A_6

(D) None of these

2) If switch S is closed, the ammeters that read less than 3A

(A) A_1

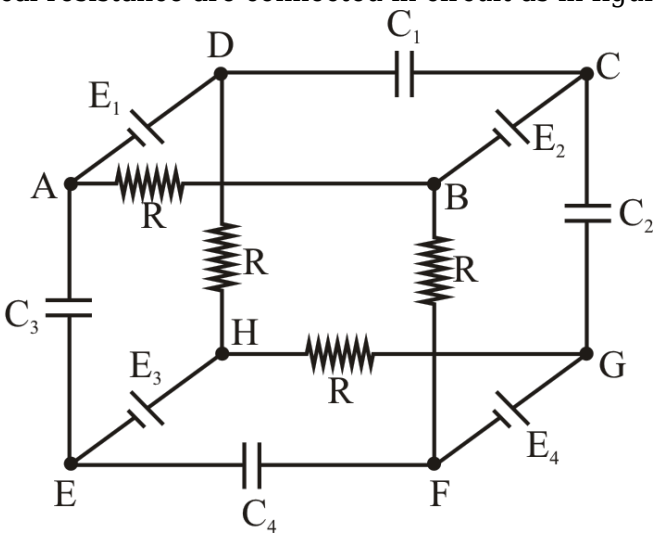
(B) A_2

(C) A_3

(D) A_4

Common Content for Question No. 3 to 4

Four batteries of emf E_1 = 4V, E_2 = 8V, E_3 = 12 V and E_4 = 16 V and four identical capacitors each of 1 μF and four identical resistance are connected in circuit as in figure. Batteries do not have any internal resistances



3) At steady state potential difference across C_2 is :

(A) 1V

(B) 5V

(C) 4V

(D) 6V

4)

If points H and B are shorted, the new charge on C_2 at steady state will be :

(A) Zero

(B) 5 µC

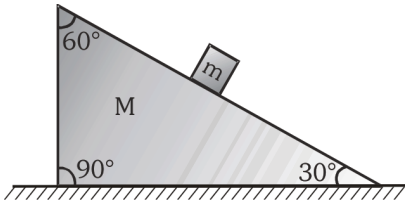
(C) 4 µC

(D) 1µC

SECTION-II

1)

A triangular block of mass M with angle 30°, 60°, 90° rests with its 30° – 90° side on a horizontal smooth fixed table. A cubical block of mass m rests on the 60° – 30° side of the tringular block. What horizontal acceleration (in m/s^2) a must M have relative to the stationary table so that m remains stationary with respect to the triangular block- [M = 9 kg, m = 1 kg, $\sqrt{3}$ = 1.73, g = 9.8 m/s^2]



2) A man of mass 75 kg is pushing a heavy box on a flat floor. The coefficient of kinetic and static friction between the floor and the box is 0.20, and the coefficient of static friction between the man's shoes and the floor is 0.80. If the man pushes horizontally, if the maximum mass (in kg) of the box he can slip is 100x write value of x ?

3) Figure (i) shows a block B of mass 4 kg kept on a rough ground, with a block A of mass 2 kg kept upon it. When a horizontal force F = 32 N is applied to block B, the relative acceleration of B w.r.t. A is shown in figure (ii) below, plotted against time. If the roughness coefficient (µ) between the surfaces of A and B and that between B and ground be the same, find the value of [10 µ] where [*] means greatest integer function. Fig. (i)

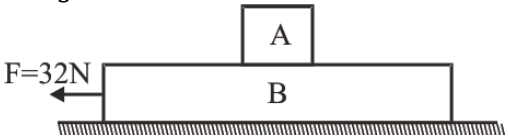
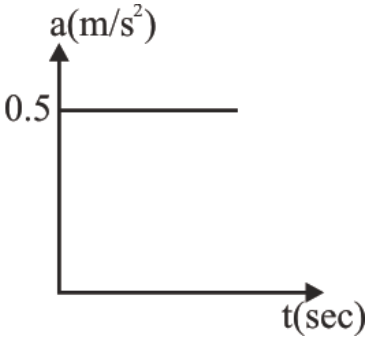
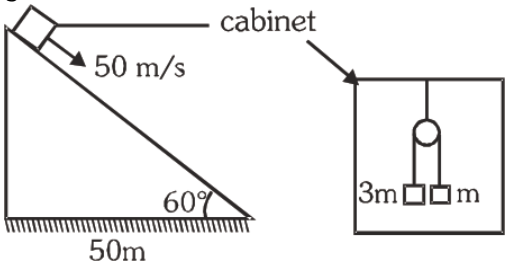


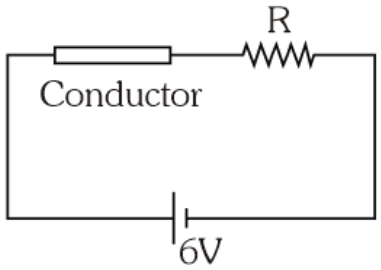
Fig. (ii)



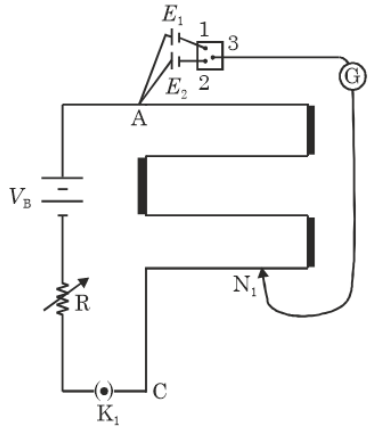
4) A cabinet is sliding on the inclined surface with constant velocity as shown in figure. In cabinet there is a block pulley system moving with velocity same as cabinate. The speed of block having greater mass w.r.t. cabinet is 5α m/s after 2s of its release. Find the value of α.



5) The current developed in a real conductor is given as $I = aV^{1/2}$, where V is the potential difference across the conductor. When it is connected with a resistance of R Ω in series with a battery of 6 Volt as shown in the figure, the power developed in the conductor is twice the power generated in resistance of RΩ. Find the value of R. [$a = 0.2A\ Volt^{-01/2}$]



6) In an experiment with a potentiometer, $V_B = 10V$, R is adjusted to be 50 Ω. A student wanting to measure voltage E_1 of a battery (approx. 8V) finds no null point possible. He then diminshes R to 10Ω and is able to locate the null point on the last (4th) segment of the potentiometer. The resistance of the potentiometer wire (R') will be a < R' < b. Find the value of b (in Ω).



7) A particle is projected radially outwards from the surface of a planet with an initial speed of 4.0 km/s. Maximum height attained by the particle is h = 900 km from the planet's surface. Find the radius R (in km) of the planet taking g = 10 m/s^2 on the surface of the planet. Fill R/900.

8) Two particles of combined mass M, placed in space with certain separation, are released. Interaction between the particles is only of gravitational in nature and there is no external force present. Acceleration of one particle with respect to each other when separation between them is R, has a magnitude $\frac{xGM}{R^2}$. Value of x is

PART-2 : CHEMISTRY

SECTION-I (i)

1)

1.22 gm C_6H_5COOH is added into 100 gm of acetone, elevation in B.P. is found to be 0.17. If K_b (acetone) = 1.7 K - Kg mol^{-1} then select the incorrect options

(A) Observed molar mass of benzoic acid is 61 gm/mole.

(B) Observed molar mass of benzoic acid is 122 gm/mole

(C) Benzoic acid neither associate nor dissociate.

(D) Benzoic acid exist as dimer in acetone due to intermolecular H-bonding

2) In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) :

(A) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

(B) The activation energy of the reaction is unaffected by the value of the steric factor

(C) Since P = 4.5, the reaction will not proceed unless an effective catalyst is used.

(D) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation.

3) The standard reduction potentials of some half cell reactions are given below:

$PbO_2 + 4H^+ + 2e^- \rightleftharpoons Pb^{2+} + 2H_2O$; $E^0 = 1.45V$

$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$; $E^0 = 1.51\ V$

$Ce^{4+} + e^- \rightleftharpoons Ce^{3+}$; $E^0 = 1.61\ V$

$H_2O_2 + 2H^+ + 2e^- \rightleftharpoons 2H_2O$; $E^0 = 1.71\ V$

Pick out the correct statement (for standard condition of all species) :

(A) Ce^{4+} will oxidise Pb^{2+} to PbO_2

(B) MnO_4^- will oxidise Pb^{2+} to PbO_2

(C) H_2O_2 will oxidise Mn^{2+} to MnO_4^-

(D) PbO_2 will oxidise Mn^{2+} to MnO_4^-

4)

Which of the following statement is/are correct

(A) The peroxide ion has a bond order of 1 while the oxygen molecule has a bond order of 2

(B) The peroxide ion has a weaker bond than the dioxygen molecule has.

(C) The peroxide ion as well as the dioxygen molecules are paramagnetic

(D) The bond length of the peroxide ion is greater than that of the dioxygen molecule

5) How many of the following trend is "**INCORRECT**" ?

(A) Electron affinity : Se > S > Br > C□

(B) Ionic radii : $Ba^{2+} > Rb^+ > K^+ > Sr^{2+}$

(C) Atomic radii : S > C□ > O > F

(D) $I.E_2$ (Be) > $I.E_2$ (Li)

6) When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A (expressed in eV) and de–Broglie wavelength λ_A. The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.20 V is $T_B = T_A$ –1.50 eV. If the de–Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$ then :

(A) the work function of A is 2.25 eV

(B) the work function of B is 3.70 eV

(C) T_A = 2.00 eV

(D) T_B = 2.75 eV

SECTION-I (ii)

Common Content for Question No. 1 to 2

Dipole moment of a molecule is a vector sum of all the individual bond moments. Mathematically it is $\mu = q \times d$, where q is magnitude of particle charges and d is the distance between centre of opposite charges. Its main application includes, determination of polarity of bond, symmetry of molecule and in calculation of % ionic character.

1) In HI molecule if observed dipole moment in 1.2 Debye and also the interionic distance between atoms (d_{HI}) is 1Å. The percentage covalent character in HI bond will be

(A) 25%

(B) 75%

(C) 50%

(D) 12.5%

Common Content for Question No. 2 to 3

For the first order dissociation reaction

$A_2B_3(aq.) \rightarrow 2A^{3+}(aq.) + 3B^{2-}(aq.)$

a definite volume of solution is taken at different instant and its osmotic pressure rise is measured. If osmotic pressure at t = 0 is 2 mm of Hg and at t = 10 min is 6 mm of Hg .
[Given density of solution is constant and equal to 1gm/ml]

2) Osmotic pressure of solution in mm of Hg at t = 20 min is :

(A) 7 mm

(B) 8 mm

(C) 10 mm

(D) 12 mm

Common Content for Question No. 3 to 4
The electrochemical cell shown below is a concentration cell.
M | M^{2+} (saturated solution of a sparingly soluble salt, MX_2) | | M^{2+} (0.001 mol dm^{-3}) | M
The emf of the cell depends on the difference in concentrations of M^{2+} ions at the two electrodes.
The emf of the cell at 298 K is 0.059V.

3) The solubility product (K_{sp} ; mol^3 dm^{-9}) of MX_2 at 298 K based on the information available for the given concentration cell is $x \times 10^{-15}$ (take $2.303 \times R \times 298/F = 0.059$ V). The value of x is ?

(A) 1×10^{-15}

(B) 4×10^{-15}

(C) 1×10^{-12}

(D) 4×10^{-12}

Common Content for Question No. 4 to 5
An ideal solution is obtained by mixing a non-volatile solute B with a volatile solvent A (molar mass = 60). If the mass ratio of A : B in solution is 10 : 1 and vapour pressure of pure A is 400 mm and vapour pressure decreases by 4% on forming the above solution at 300K.

4) The molality the solution is -

(A) 1

(B) $\frac{36}{25}$

(C) $\frac{25}{36}$

(D) $\frac{36}{25}$

SECTION-II

1) In the reaction : N_2H_4 (□) + ClO_3^- (aq.) → NO(g) + Cl^- (aq.) number of moles of electrons gained by 1.5 moles of ClO_3^- (aq.) will be :-

2)

19 gm mixture of Na_2CO_3(s) and $NaHCO_3$(s) on heating gives 2.2 gm CO_2 gas.

Find % $NaHCO_3$ (by mass) in mixture.

3) The decomposition reaction $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After $Y \times 10^3$ s, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is 5×10^{-4} s^{-1}, assuming ideal gas behavior, the value of Y is ___

4) A current of 9.65 micro-amperes is passed through $AuCl_3$ solution for 30 minutes using Pt-electrodes causing deposition of a uniform single atom thick layer of total area 600 cm^2. Then average surface area covered by a single gold atom is 1.66×10^{-b} cm^2 then find the value of b. ($N_A = 6 \times 10^{23}$).

5) An aqueous solution containing 28% by mass of liquid (mol. mass = 140) has a vapour pressure of 160 mm at 30 °C. Find the vapour pressure of the pure liquid A. (The vapour pressure of water at 30 °C is 150 mm)

6) The bond length in HCl molecule is 0.127 nm. The observed dipole moment of HCl is 1.08 D. The percentage ionic character in HCl molecule is_____.

7) The uncertainty in the measurement of position of an electron in 2nd excited state of He^+ is $\frac{9}{16}$ Å. What will be the minimum uncertainty in the measurement of De-Broglie wave length of this electron in angstorms(Å) ? [take first Bohr radius = 0.5 Å] [π = 3.14]

8) Zn Amalgam is prepared by electrolysis of aqueous $ZnCl_2$ using Hg cathode (9gm). How much current (in ampere) is to be passed through $ZnCl_2$ solution for 1000 seconds to prepare a Zn Amalgam with 25% Zn by wt.
(Zn = 65.4)

PART-3 : MATHEMATICS

SECTION-I (i)

1) Let $N = \frac{\log_2 24}{\log_{96} 2} - \frac{\log_2 192}{\log_{12} 2}$. Then N is

(A) a natural number

(B) a prime number

(C) a rational number

(D) an integer

2) In a survey, 100 teenagers were interviewed to know their preferences for arrange marriage

(AM), love marriage (LM), remain single (RS), and the collected data were as following
n(AM ∩ LM ∩ RS) = 10, n(AM ∩ LM) = 20
n(LM ∩ RS) = 30, n(AM ∩ RS) = 25
n(only AM) = 25, n(only LM) = 5, n(only RS)=8 and some of them did not respond,
then correct statement(s) from the following is/are

(A) n(AM) = 47

(B) n(LM) = 45

(C) n(RS) = 53

(D) $n(AM \cup LM \cup RS) = 100$

3) Given $f(1) = 2$ and $f(n+1) = \frac{f(n)-1}{f(n)+1} \forall n \in N$, then which of the following option is/are correct ?

(A) $f(2015) = -\frac{1}{2}$

(B) $\left(f(2012)\right)^{f(2013)} = 9$

(C) $f(1001) = 2$

(D) $f(2015) = -3$

4) Let $f : A \to B$ and $g : B \to C$ be two functions such that $gof : A \to C$ is defined. Which of the following statement(s) is/are INCORRECT?

(A) If gof is into then f must be into.

(B) If f is into and g is onto then gof must be onto.

(C) If gof is one-one then g must be one-one.

(D) If f is surjective and g is injective then gof must be surjective.

5) If the set of value of 'a' for which the equation $2\cos^{-1} x = a + a^2 (\cos^{-1} x)^{-1}$ possess a solution is $[k_1\pi, k_2\pi]$ then

(A) $k_2 - k_1$ = prime number

(B) Number of integral values of 'a' are 10

(C) Sum of all integral values of 'a' is less than (– 10)

(D) $|k_1 - k_2|$ is odd Prime Number

6) If $t_1 = \left(\sin^{-1} x\right)^{\sin^{-1} x}$; $t_2 = \left(\sin^{-1} x\right)^{\cos^{-1} x}$; $t_3 = \left(\cos^{-1} x\right)^{\sin^{-1} x}$; $t_4 = \left(\cos^{-1} x\right)^{\cos^{-1} x}$, then which among the following option(s) is/are CORRECT ? (where sgn(x) represents signum function)

(A) If $x \in (0, \cos 1)$ then $\text{sgn}(t_4 - t_2 + t_3 - t_1) = 1$

(B) If $x \in \left(\cos 1, \frac{1}{\sqrt{2}}\right)$ then $\log_{10}(t_3 - t_1 + t_4 - t_2)$ is defined.

(C) If $x \in (\sin 1, 1)$ then $t_1 > t_2 > t_4 > t_3$

(D) If $x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right)$ then $t_2 > t_1 > t_3 > t_4$

SECTION-I (ii)

Common Content for Question No. 1 to 2

Let $f(x) = \begin{cases} \lim\limits_{n\to\infty} \frac{|x+1|^n + x^2}{|x| + x^{2n}} & -6 \leqslant x < 0 \\ \{\sin x\} & 0 \leqslant x < 6 \end{cases}$, where {.} denotes fractional part function.
on the basis of above information, answer the following questions :

1)

Number of solutions of the equation $f(x) = \frac{x}{6}$ in [-6, 6]

(A) 2

(B) 3

(C) 4

(D) 5

Common Content for Question No. 2 to 3

Let $f(x) = \begin{cases} \lim\limits_{n\to\infty} \frac{|x+1|^n + x^2}{|x| + x^{2n}} & -6 \leqslant x < 0 \\ \{\sin x\} & 0 \leqslant x < 6 \end{cases}$, where {.} denotes fractional part function.
on the basis of above information, answer the following questions

2) Number of points at which f(x) is not differentiable in [-6, 6]

(A) 6

(B) 5

(C) 4

(D) 3

Common Content for Question No. 3 to 4

Let p, q, r be positive integers such that $\frac{r}{q}$ is an integer. Also p, q, r (taken in that order) are in geometric progression and the arithmetic mean of p – 2, q + 5, r + 4 is (2q – p – 1).

3)

The value of p is equal to

(A) 3

(B) 4

(C) 5

(D) 6

Common Content for Question No. 4 to 5

Let p, q, r be positive integers such that $\frac{r}{q}$ is an integer. Also p, q, r (taken in that order) are in geometric progression and the arithmetic mean of p – 2, q + 5, r + 4 is (2q – p – 1).

4) The number of ordered triplet (p, q, r) is equal to

(A) 1

(B) 2

(C) 3

(D) 4

SECTION-II

1)

Let $a_1, a_2, \ldots\ldots a_n$, be real numbers such that

$$\sqrt{a_1}+\sqrt{a_2-1}+\sqrt{a_3-2}+\ldots\ldots+\sqrt{a_n-(n-1)}=\frac{1}{2}(a_1+a_2+\ldots.+a_n)-\frac{n(n-3)}{4}$$

then find the value of $\sum_{i=1}^{100} a_i$

2) The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2-11x+\alpha = 0$ are rational numbers is

3) The real function $f(x)$ has the property that whenever a, b, x are positive integers such that $a + b = 2^x$, the equation $f(a) + f(b) = x^2$ holds, then $\frac{f(2014)}{12}$ is

4) Let f(x) be a function such that $f(x-1)+f(x+1) = \sqrt{2}\, f(x)$ for all $x \in R$. If $f(3) = 5$ then $\sum_{r=0}^{10} f(3+8r)$ is equal to

5) If $a_1 = 1$ and $a_n = n(1 + a_{n-1})\ \forall\ n \geq 2$ then $\left(\lim_{n\to\infty}\left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right)\ldots\ldots\left(1+\frac{1}{a_n}\right)\right).\frac{1}{e} =$

6) Let $P_n=\prod_{k=2}^{n}\left(1-\frac{1}{{}^{k+1}C_2}\right)$. If $\lim_{n\to\infty} P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then value of $(a+b)$ is

7) If $f(x) = \{x\} + \{2x\} + \{3x\}$ is not differentiable at 7 points in (0, n), (where, $n \in N$ and $\{.\}$ is

fractional part function) then n is equal to

8) For every integer n, let a_n, b_n and c_n are real numbers.
Let continuous function f : R → R is given by

$$f(x) = \begin{cases} a_n + \cos \pi x & , \quad x \in \left[2n, \frac{6n+2}{3}\right] \\ b_n + \sin \frac{\pi x}{2} & , \quad x \in \left(\frac{4n-9}{2}, 2n-4\right) \\ \cos \frac{\pi x}{2} + \sin \pi x + c_n & , \quad x \in \left(\frac{6n+14}{3}, \frac{4n+11}{2}\right] \end{cases}$$ for all integers n

If greatest value of $a_n - a_{n+1} = \frac{\alpha + \sqrt{\beta} + \sqrt{\delta}}{\gamma}$ where β, δ are coprime number and α, β, γ, δ ∈ N, then minimum value of (α + β + γ + δ) is

ANSWER KEYS

PART-1 : PHYSICS

SECTION-I (i)

Q.	1	2	3	4	5	6
A.	B,C,D	B,C	B,D	B,C	A,B,D	A,B,D

SECTION-I (ii)

Q.	7	8	9	10
A.	D	B	B	A

SECTION-II

Q.	11	12	13	14	15	16	17	18
A.	5.66	3.00	2.00	2.00	5.00	200.00	8.00	1.00

PART-2 : CHEMISTRY

SECTION-I (i)

Q.	19	20	21	22	23	24
A.	A,D	B,D	A,B,C	A,B,D	A,B,D	A,B,C

SECTION-I (ii)

Q.	25	26	27	28
A.	B	B	B	C

SECTION-II

Q.	29	30	31	32	33	34	35	36
A.	9.00	44.21	2.30 to 2.31	14.00	360.00 to 360.20	17.69 to 17.71	3.14	8.80 to 8.90

PART-3 : MATHEMATICS

SECTION-I (i)

Q.	37	38	39	40	41	42
A.	A,B,C,D	B,C	A,B,C	A,B,C,D	A,B,C,D	A,B,C

SECTION-I (ii)

Q.	43	44	45	46
A.	B	C	C	B

SECTION-II

Q.	47	48	49	50	51	52	53	54
A.	**5050.00**	**3.00**	**9.00**	**55.00**	**1.00**	**4.00**	**2.00**	**19.00**

SOLUTIONS

PART-1 : PHYSICS

1)

Component of velocity in vertical direction in same (H_{max} is same)

$v_1 \sin\theta_1 = v_2 \sin\theta_2$ ....(1)

(A) Time period

$$T_A = \frac{2u_1 \sin\theta_1}{9}$$

$$T_B = \frac{2u_2 \sin\theta_2}{9}$$

$T_A = T_B$

(B) $$R_A \Rightarrow \frac{2u_1^2 \sin\theta, \cos\theta_1}{9}$$

$$R_B \Rightarrow \frac{2u_2^2 \sin\theta_2, \cos\theta_2}{9}$$

$R_B > R_A$

from eq^n – 1

$u_1 \cos\theta_1 < u_2 \cos\theta_2$

since H_{max} is same

$u_1 \sin\theta_1 = u_2 \sin\theta_2$

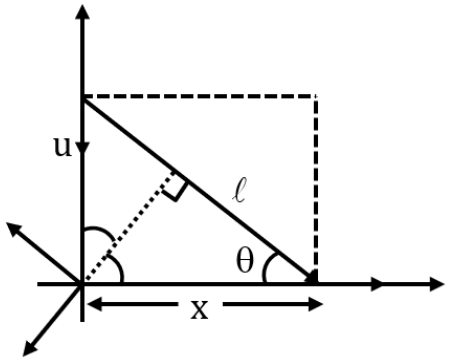
2) u = 4 m/s

t = 0.4 sec

$$t = \sqrt{\frac{2h}{g}}$$

$$h = \frac{t^2 g}{2} = \frac{0.16 \times 10}{2} = 0.8\text{m}$$

R = u × T = 4 × .4 = 1.6



3)

$$E_{\perp} = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$

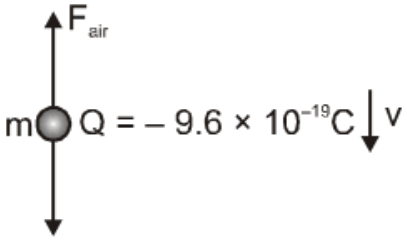
$$E_{||} = \frac{k\lambda}{r}(\cos\theta_2 - \cos\theta_1)$$

$$E_{\perp} = \frac{K\lambda}{x \sin\theta}(\sin\theta - \cos\theta)$$

$$E_{||} = \frac{K\lambda}{x \sin\theta}(\sin\theta + \cos\theta)$$

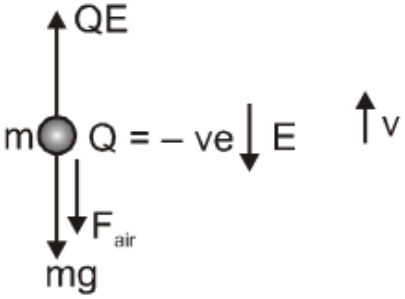
$$E_{net} = \left(\frac{2\sqrt{2}k\lambda}{2\ell \sin\theta \cos\theta}\right)$$

$$E = \frac{2\sqrt{2}K\lambda}{\ell \sin 2\theta}$$



4) (initially)

 $mg = f_{air}$



(finally)

 $QE = mg + f_{air} = 2mg$

 charge is –ve, so electric field 'E' is directed downwards.

& QE = 2 mg

 $$E = \frac{2mg}{Q} = \frac{2 \times 1.6 \times 10^{-18} \times 10}{9.6 \times 10^{-19}} = \frac{1}{3} \times 10^2 NC^{-1}$$

5) $$qE2a = \frac{1}{2}m\left(4v^2 - v^2\right)$$

$$qE2a = \frac{3}{2}mv^2$$

$$E = \frac{3}{4}\frac{mv^2}{qa}$$

$$P_p = q\frac{3}{4}\frac{mv^2}{qa}v = \frac{3}{4}\frac{mv^3}{a}$$

$P_Q = 0$

6) $$F_B = q\left(\vec{v} \times \vec{B}\right) R = \frac{mv_{\perp}}{qB}$$

Both moves in helical because of two comp.

$V_{\perp}$ & $V_{||}$

$R \propto V_{\perp}$

$\therefore R_1 > R_2$

Pitch $\propto V_{||} \therefore P_2 > P_1$

$\therefore T = \frac{2\pi m}{qm}$

$\therefore$ Time period same.

7)

Use KCL

Before connection

$A_1 = A_2 = 9.5$ A, $A_4 = A_5 = 5$A, $A_3 = A_6 = 2$

8)

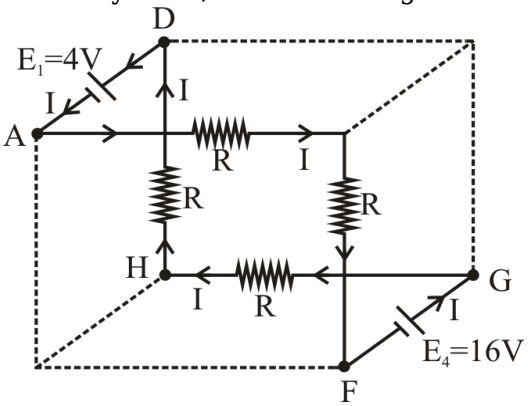
After connection

$A_1 = 12.5$ A, $A_2 = 2.5$ A, $A_3 = 10$A,

$A_4 = 7$A, $A_5 = 8$A, $A_6 = 5$A

9)

In steady state, current through branch of capacitor is zero,



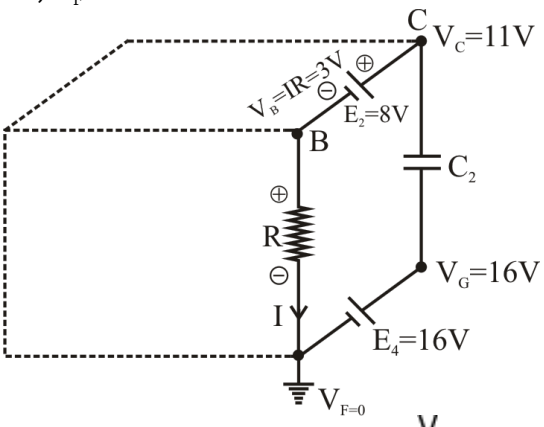
Current will flow as shown in the diagram.

$I = \frac{16-4}{4R} = \frac{3}{R}$

For potential difference across C_2

Let, F be the reference position,

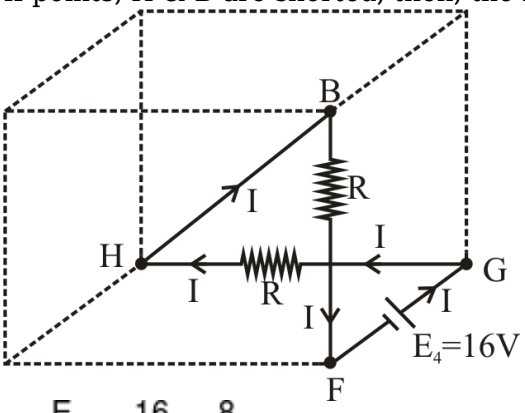
So, $V_F = 0$



Potential difference across C_2, V_{C_2} = 5V

10)

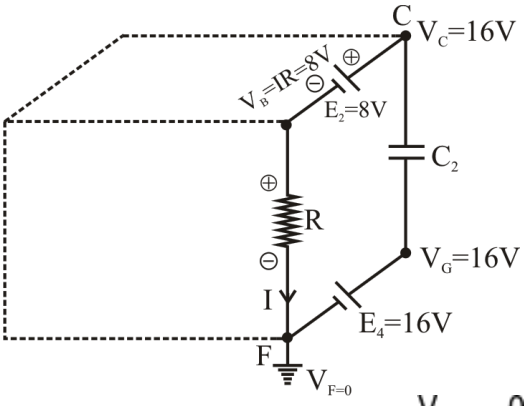
If points, H & B are shorted, then, the new path of current I is :



$$I = \frac{E_4}{2R} = \frac{16}{2R} = \frac{8}{R}$$

For charge on C_2, let F be the reference position,

So, $V_F = 0$



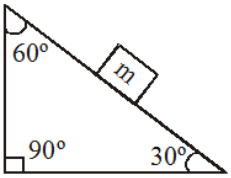
Potential difference across C_2, $V_{C_2} = 0$

So, charge on $C_2 = 0$

11) M = 9kg

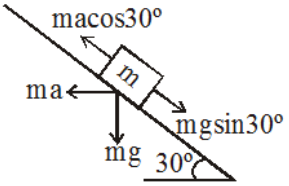
m = 1kg

g = 9.8 m/s^2



Let acceleration 'a' is given to 'M'

So block 'm' get for "ma" as

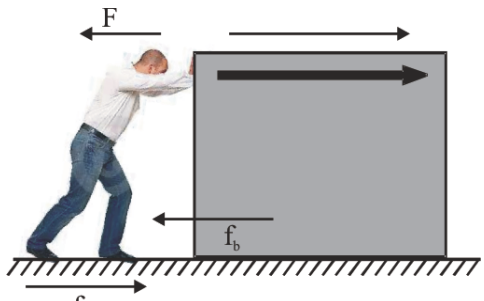


for constant velocity of zero of block 'm'

ma cos 30º = mg sin 30º

a = g tan 30º

$\boxed{a = 5.658 \text{ m/s}^2}$



12)

$F > f_b = 0.2\ mg$

$0.2\ mg < F < \mu_S \times 75g$

$m < 75 \times 4 = 300\ kg$

13) $32 - 60\mu - 20\mu = 4a_B$ .....(i)

$20\mu = 2a_A$....(ii)

$a_B - a_A = 8 - 20\mu - 10\mu = 0.5 \Rightarrow \mu = 0.25$

14) $g_{eff.} = g$

So speed of Heavier Block w.r.t lift = at

$= \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g \times t = 10$

15) $I = aV^{1/2} = \frac{6 - V}{R}$ ..... (i)

$P_{con} = aV^{3/2}, P_R = \frac{(6-V)^2}{R}$

According question $P_{con} = 2P_R$

$\frac{2(6-V)^2}{R} = aV^{3/2}$

Divide (ii) by (i)

$V = 2(6 - V)$

$\Rightarrow 3V = 12 \Rightarrow V = 4$ Volt

$2a = \frac{2}{R} \Rightarrow R = \frac{1}{a} = 5\Omega$

16) Let R' be the resistance of the potentiometer wire.

$\frac{10 \times R'}{50 + R'} < 8 \Rightarrow R' < 200\Omega$

$\frac{10 \times R'}{10 + R'} > 8 \Rightarrow R' > 40$

From above we can write, . Hence a = 40 and b = 200

$\therefore \left(\frac{b}{a}\right) = 5$

17) $\frac{1}{2}mv^2 - \frac{GMm}{R} = -\frac{GMm}{R+h}$

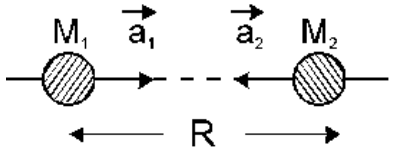
$$\frac{v^2}{2} = \frac{GMh}{R(R+h)} = \frac{GM}{R^2}\frac{Rh}{R+h}$$

$$\frac{v^2}{2} = g \times \frac{Rh}{R+h}$$

$(R + 900 \times 10^3) \times 8 \times 10^6 = 10 \times R \times 900 \times 10^3$

$R = 72 \times 10^5$ m $\Rightarrow 7200$ km $\Rightarrow \frac{R}{900} = 8$

18)



$a_1 = \frac{GM_1M_2}{R^2} / M_1$

$a_2 = \frac{GM_1M_2}{R^2} / M_2$

acceleation of M_1 w.r.t. M_2

$a_{rel.} = a_1 + a_2$

$= \frac{G(M_1 + M_2)}{R^2} = \frac{GM}{R^2}$

PART-2 : CHEMISTRY

19)

$\Delta T_b = i.K_b.m$

$0.17 = i \times 1.7 \times \frac{1.22/122}{100} \times 1000$

i = 1, so no association or dissociation take place.

20) $P = \frac{K_{actual}}{K_{theoretical}}$

P = 4.5

$K_{actual} > K_{theoretical}$a

$K_{actual} = K_{exp.} = PAe^{-E_a/RT}$

$A_{actual} > A_{theorectical}$

Hence experimental determined value of frequency factor is higher that predicted by Arrhenius equation.

Activation energy of reaction is unaffected by the value of the steric factor.

Since P = 4.5 P > 1 so no catalyst is required.

21)

A) $E^\circ_{Ce^{4+}} > E^\circ_{Pb^{4+}}$ thus Ce^{4+} will be reduced.

B) $E^{\circ}_{Mn^{7+}} > E^{\circ}_{Pb^{4+}}$ thus MnO_4^- will be reduced.
C) $E^{\circ}_{H_2O_2} > E^{\circ}_{MnO_4^-}$ thus H_2O_2 will be reduced.
D) $E^{\circ}_{Pb^{4+}} < E^{\circ}_{MnO_4^-}$ thus it means that MnO_4^- will be reduced and hence Pb^{2+} will be oxidized to Pb^{4+}.
Hence options (A), (B) and (C) are correct.

22)

Peroxide ion = $O_2^{2\ominus}$ ⇒ B.O. = 1
O_2 molecule ⇒ B.O. = 2
B.O. ∝ B.E. ∝ $\frac{1}{B.L.}$
Correct option is A,B,D.

23)

(B) G-1 and G-2 Ionicradii

G-1	G-2
3	1(smallest)
5	2
8	4
9	6
10(Highest)	7

(D) $I.E_3$ (Li) > $I.E_2$ (He)

24) $T_A = 4.25 - W_A$
$T_B = 4.20 - W_B$

$$\lambda_A = \sqrt{\frac{150}{T_A}}$$

$$\lambda_B = \sqrt{\frac{150}{T_B}}$$

$$2\lambda_A = \sqrt{\frac{150}{T_A - 1.50}}$$

T_A = 2 eV, T_B = 0.5 eV, W_A = 2.25 eV, W_B = 3.7 eV

25) % ionic = $\frac{\mu_{obs}}{\mu_{theo}} \times 100 = \frac{1.2 \times 10^{-18} esu\ cm \times 100}{4.8 \times 10^{-10} esu \times 10^{-8} cm} = 25\%$
⇒ % covalent = 100 – 25 = 75%

26) $A_2B_3 \rightleftharpoons 2A^{3+}_{(aq.)} + 3B^{-2}_{(aq.)}$
t = 0 a_0
t = 10 $a_0 - x$ 2x 3x
π ∝ C , where C effective molarity
$\pi_0 \propto a_0$

$\pi \propto (a_0 + 4x)$

$x \propto [(\pi - a_0)/4]$

$$K = \frac{1}{t}\ell n\left(\frac{a_0}{a_0 - x}\right)$$

$$= \frac{1}{t}\ell n\frac{a_0}{a_0 - \left(\frac{\pi - a_0}{4}\right)} \Rightarrow \frac{1}{t}\ell n\left[\frac{4a_0}{5a_0 - \pi}\right]$$

$$K = \frac{1}{10}\ell n\left(\frac{4 \times 2}{5 \times 2 - 6}\right) \Rightarrow \frac{1}{10}\ell n2$$

$$t_2 = \frac{1}{K}\ell n\left(\frac{4 \times 2}{5 \times 2 - x'}\right)$$

$$20 = \frac{10}{\ln 2}\ln\left(\frac{8}{10 - x'}\right)$$

$\boxed{x' = 8\text{mm Hg}}$

27) $0.059 = \dfrac{-0.059}{2} \log \dfrac{[M^{2+}]_A}{0.001}$

or, $\dfrac{[M^{2+}]_A}{0.001} = \dfrac{1}{100}$ or, $[M^{2+}]_A = 10^{-5} = s$

$MX_2(s) \rightleftharpoons M^{2+}(aq) + 2X^-(aq)$

s 2s

$Ksp = 4s^3 = 4 \times 10^{-15}$

28) $\dfrac{\Delta P}{P^0} = 0.04 = X_B$, $m = \dfrac{0.04 \times 1000}{0.96 \times 60} \Rightarrow \dfrac{25}{36}$

$$\overset{-2}{N_2}\overset{+1}{H_4}(\ell) + \overset{+5}{ClO_3^-}(aq) \rightarrow \overset{+2\ -2}{NO}(g) + Cl^{-1}(aq)$$

Reduction

29)

n_f of $ClO_3^{-1} = 6$

$\Rightarrow$ 1 mole of ClO_3^{-1} gained $\rightarrow 6e^-$

$\therefore$ 1.5 mole of ClO_3^{-1} gained $\rightarrow (6 \times 1.5)\ e^-$

$= 9\ e^-$

30)

$$Na_2CO_3 \xrightarrow{\Delta} \text{No decomposition}$$

$$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$$

$$n_{CO_2} = \frac{2.2}{44} = 0.05$$

$n_{NaHCO_3} = 2 \times 0.05 = 0.1$

Wt. of $NaHCO_3 = 0.1 \times 84 = 8.4$ gm

% $NaHCO_3$ in sample $= \dfrac{8.4}{19} \times 100 = =$ **44.21**

31) $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ at constant V, T

t = 0 1

t = y × 10^3 sec (1 – 2P) 2P P

$P_T = (1 + P) = 1.45$

P = 0.45 atm

$(2K)t = \dfrac{2.303 \log\left(\dfrac{1}{1-2P}\right)}{}$

$(2 \times 5 \times 10^{-4}) \times y \times 10^3 = \dfrac{2.303 \log \dfrac{1}{0.1}}{}$

y = 2.303 = 2.30

$$n_F = \frac{9.65 \times 10^{-6} \times 30 \times 60}{96500}$$

$$= 18 \times 10^{-8}$$

$$Au^{3+} + 3e \rightarrow Au$$

$$18 \times 10^{-8} \qquad 6 \times 10^{-8}\ mol$$

$$(6 \times 10^{-8} N_A)\ Au\ atoms\ have\ area = 600\ cm^2$$

$$\text{Area of 1 Au - Atom} = \frac{600}{6 \times 10^{-8} \times N_A}$$

$$= 1.66 \times 10^{-14}\ cm^2$$

32)

33) For two miscible liquids,

P_{total} = Mole fraction of A × P_A^0 + Mole fraction of B × P_A^0

Number of moles of $A = \dfrac{28}{140} = 2$

Liquid B is water. Its mass is (100 – 28), i.e., 72.

Number of moles of $B = \dfrac{72}{18} = 4.0$

Total number of moles = 0.2 + 4.0 = 4.2

Given, P_{total} = 160 mm

p_B^0 = 150 mm

So, $160 = \dfrac{0.2}{4.2} \times P_A^0 + \dfrac{4.0}{4.2} \times 150$; $P_A^0 = \dfrac{17.15 \times 4.2}{0.2} = 360.15 mm$

34)

$\mu_{theoritical}$ = q × d

$$= \frac{1.6 \times 10^{-19} \times 0.127 \times 10^{-9}}{3.33 \times 10^{-30}} = 6.10\,D$$

$\mu_{observed}$ = 1.08 D

% Ionic character = $\dfrac{1.08}{6.10} \times 100 = 17.70\%$

35) $\lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$

$\frac{dp}{d\lambda} = -\frac{h}{\lambda^2}$

$\therefore \Delta p = \frac{h}{\lambda^2} \cdot \Delta\lambda$

$\Rightarrow \Delta x \cdot \Delta p \geqslant \frac{h}{4\pi}$

$\Rightarrow \Delta x \cdot \frac{h}{\lambda^2} \cdot \Delta\lambda \geqslant \frac{h}{4\pi}$

$\Rightarrow \Delta x \cdot \Delta\lambda \geqslant \frac{\lambda^2}{4\pi}$ ..... (1)

n = 3, z = 2

$2\pi \cdot r_0 \times \frac{9}{2} = 3 \cdot \lambda$

λ = 1.5 πÅ

$\Delta x \cdot \Delta\lambda \geqslant \frac{9}{4}\frac{\pi^2}{4\pi} \Rightarrow \Delta\lambda \geqslant \pi$

36) Let x gm of Zn deposit on 9 gm of Hg

% of Zn in Amalgam

$= \frac{x}{9+x} \times 100 = 25$

 x = 3gm

Eq. of Zn = $\frac{3 \times 2}{65.4}$

Current = $\frac{6}{65.4} \times \frac{96500}{1000}$ = 8.85 amp.

PART-3 : MATHEMATICS

37) $N = \log_2 24 \times \log_2 96 - \log_2 192 \times \log_2 12$

$= (\log_2 12 + \log_2 2) \times (\log_2 12 + \log_2 8) - (\log_2 12 + \log_2 16) \times (\log_2 12)$

put $\log_2 12 = a$

N = (a + 1) (a + 3) – (a + 4)a

= 3

38) $n(A \cup B \cup C) = n(A) + n(B) + n(C)$

$-n(A \cap B) - n(B \cap C) - n(C \cap A)$

$+ n(A \cap B \cap C)$

39)

$f(n+1) = \frac{f(n)-1}{f(n)+1} = \frac{\frac{f(n-1)-1}{f(n-1)+1} - 1}{\frac{f(n-1)-1}{f(n-1)+1} + 1}$

$= \frac{-1}{f(n-1)}$

$f(n+1) = -\frac{1}{f(n-1)}$ .....(i)

In a similar way $f(n-1) = -\frac{1}{f(n-3)}$ .....(ii)

from (i) and (ii)

$$f(n+1) = -\frac{1}{\frac{-1}{f(n-3)}} = f(n-3)$$

or $f(n + 4) = f(n)\ \forall\ n \in N$

hence $f(n)$ is a periodic sequence with period 4

using (A) put n = 1, 2, 3

we get $f(2) = \frac{1}{3},\ f(3) = -\frac{1}{2},\ f(4) = -3$

Now $f(2012) = f(4) = -3$

$f(2013) = f(1) = 2$

$f(2015) = f(3) = -\frac{1}{2}$

$f(1001) = f(1) = 2$

Consider $f: R \to R,\ f(x) = (2x)$

$g: R \to R,\ g(x) = \sin(x)$

Thus, $g \circ f: R \to R;\quad g(f(x)) = \sin(2x)$

Here gof is into but f in onto.

Thus (A) is false.

Consider $f: R \to [0, \infty), f(x) = x^2 + 1$

$g: [0, \infty) \to [0, \infty), g(x) = \sqrt{x}$

Thus, $g \circ f: R \to [0, \infty),\quad g(f(x)) = \sqrt{x^2 + 1}$

Here f is into, g is onto but gof is into.

Thus (B) is false.

Consider $f: [1, \infty) \to [0, \infty),\quad f(x) = \sqrt{x-1}$

$g: R \to R,\quad g(x) = x^2$

Thus, $g \circ f: [1, \infty) \to R,\quad g(f(x)) = x - 1$

Here gof is one-one but g is many-one.

Consider $f: R \to R,\ f(x) = (2x)$

$g: R \to R,\ g(x) = e^x$

Thus, $g \circ f: R \to R;\quad g(f(x)) = e^{2x}$

Here f is surjective and g is injective but

40) gof is not surjective

41) $2t = a + \frac{a^2}{t}$

$2t^2 = at + a^2$

$2t^2 - at - a^2 = 0$

$$t = \frac{a \pm \sqrt{a^2 + 8a^2}}{4}$$

$$t = \frac{a \pm 3a}{4}$$

$$t = \frac{a \pm 3a}{4}$$

(1) $t = \frac{a + 3a}{4}$

$t = a$

$\cos^{-1} x = a$

$a = [0, \pi]$

(2) $t = \frac{a - 3a}{4}$

$t = \frac{-2a}{4}$

$t = \frac{-a}{2}$

$\cos^{-1} x = \frac{-a}{2}$

$0 \leqslant \frac{-a}{2} \leqslant \pi$

$-\pi \leqslant \frac{a}{2} \leqslant 0$

$-2\pi \leq a \leq 0$

42) $x \in (0, \cos 1) \Rightarrow \cos^{-1} x > 1 > \sin^{-1} x$

$x \in \left(\cos 1, \frac{1}{\sqrt{2}}\right) \Rightarrow \sin^{-1} x < \cos^{-1} x < 1$

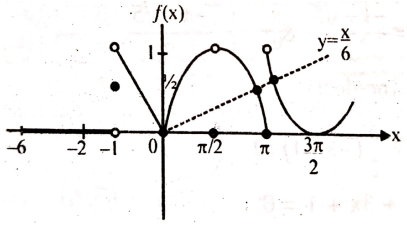
$x \in (\sin 1, 1) \Rightarrow \sin^{-1} x > 1 > \cos^{-1} x$

$x \in \left(\frac{1}{\sqrt{2}}, \sin 1\right) \Rightarrow \cos^{-1} x < \sin^{-1} x < 1$

43) $f(x) = \frac{x}{6}$ has 3 solutions.

44)

$$f(x) = \begin{cases} 0 & -6 \leq x < -1 \\ \frac{1}{2} & x = -1 \\ -x & -1 < x < 0 \\ \{\sin x\} & 0 \leq x < 6 \end{cases}$$



Not differentiable at 4 point

45)

46)

p, q, r → G.P.
A, AR, AR^2 → G.P. (R ∈ I)
Given, $\frac{A-2+AR+5+AR^2+4}{3}$ = 2AR – A – 1
$A(1 + R + R^2) + 7 = 6AR - 3A - 3 \Rightarrow A(R^2 - 5R + 4) = -10$
$\Rightarrow A(R-4)(R-1) = -10 \Rightarrow A = \frac{10}{(1-R)(R-4)} \in I^+$
Since A is a positive integer
 R = 2 or R = 3
If R = 2 and R = 3, then A = 5
 Value of p = 5
Numbers are 5, 10, 20 or 5, 15, 45
 Total number of ordered triplets of (p, q, r) are 2. Ans.

47) $a_1 + a_2 + ... + a_n - 2\sqrt{a_1} - 2\sqrt{a_2 - 1} ... - 2\sqrt{a_n - (n-1)} - \frac{n(n-3)}{2} = 0$
$(\sqrt{a_1} - 1)^2 + \left(\sqrt{a_2 - 1} - 1\right)^2 + ... + \left(\sqrt{a_n - (n-1)} - 1\right)^2 = 0$
$a_1 = 1, a_2 = 2, ... a_n = n$
$\sum_{i=1}^{100} a_i = \frac{100 \times 101}{2} = 5050$

48) $6x^2 - 11x + \alpha = 0$
For roots to be rational, D must be perfect square.
$\Rightarrow 121 - 24\alpha = \lambda^2$
⇒ maximum value of α is 5
α = 1 ⇒ λ ∉ I
α = 2 ⇒ λ ∉ I
α = 3 ⇒ λ ∈ I ⇒ 3 integral values
α = 4 ⇒ λ ∈ I
α = 5 ⇒ λ ∈ I

49) $f(a) + f(b) = x^2$ & $a + b = 2^x$
$f(a) + f(2^x - a) = x^2$
Put a = 34 & x = 11
$f(34) + f(2014) = 11^2$ ...(1)
Put a = 30 & x = 6
$f(30) + f(34) = 6^2$ ...(2)
Put a = 2 & x = 5
$f(2) + f(30) = 5^2$ ...(3)
Put a = 2 & x = 2
$f(2) + f(2) = 2^2 \Rightarrow f(2) = 2$
now equation (1) + (3) – (2)

$\Rightarrow f(2014) + f(2) = 110 \Rightarrow f(2014) = 108$

50) The given equation is
$f(x-1) + f(x+1) = \sqrt{2} f(x)$ ... (i)
Replace x by x – 1 in (i)
$f(x-2) + f(x) = \sqrt{2} f(x-1)$ ... (ii)
Replace x by x + 1 in (ii)
$f(x) + f(x+2) = \sqrt{2}\, f(x+1)$ ... (iii)
Adding (ii) and (iii), we have
$f(x-2) + f(x+2) + 2f(x) = \sqrt{2}\,(f(x-1)+f(x+1))$
$= \sqrt{2}\,\sqrt{2}\, f(x) = 2f(x)$
$\Rightarrow f(x-2) = -f(x+2)$
Replacing x by x + 2, we have
$f(x) = -f(x+4) = -(f(x+8)) = f(x+8)$

$$\sum_{r=0}^{10} f(3+8r) = 11 \times f(3) = 11 \times 5 = 55$$

51) $a_2 = 2(1 + a_1)$

$$\Rightarrow 1 + a_1 = \frac{a_2}{2}, 1 + a_2 = \frac{a_3}{3} \ldots\ldots 1 + a_{n-1} = \frac{a_n}{n}$$

Given limit =

$$\lim_{n\to\infty} \left(\frac{1+a_1}{a_1}\right)\left(\frac{1+a_2}{a_2}\right) \ldots\ldots \left(\frac{1+a_n}{a_n}\right)$$

$$= \lim_{n\to\infty} \frac{1}{2}\cdot\frac{1}{3} \ldots\ldots \frac{1}{n}(1+a_n) = \lim_{n\to\infty} \frac{(1+a_n)}{n!}$$

But $\frac{a_n}{n!} = \frac{1}{(n-1)!} + \frac{a_{n-1}}{(n-1)!}$

$$\Rightarrow \frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

Put n = 2, 3, 4 ..... $\frac{a_2}{2!} - \frac{a_1}{1!} = \frac{1}{1!}$

$$\frac{a_3}{3!} - \frac{a_2}{2!} = \frac{1}{2!}$$

$$\frac{a_n}{n!} - \frac{a_{n-1}}{(n-1)!} = \frac{1}{(n-1)!}$$

$$\frac{a_n}{n!} - \frac{a_1}{1!} = \frac{1}{1!} + \frac{1}{2!} + \ldots\ldots + \frac{1}{(n-1)!}$$

$$\therefore \frac{a_n}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \ldots\ldots + \frac{1}{(n-1)!}$$

$$\therefore \lim_{n\to\infty} \left(\frac{1+a_n}{n!}\right) = \lim_{n\to\infty} \frac{1}{n!} + \lim_{n\to\infty} \frac{a_n}{n!} = e$$

$\therefore$ Required value = $e \times \frac{1}{e} = 1$

52) $1 - \frac{1}{{}^{k+1}C_2}$

$$\Rightarrow 1-\frac{1}{\left(\frac{(k+1)k}{2!}\right)}$$

$$\Rightarrow 1-\frac{2}{(k+1)k}$$

$$\Rightarrow \frac{k^2+k-2}{k(k+1)}$$

$$\Rightarrow \frac{(k+2)(k-1)}{k(k+1)}$$

$$P_n=\prod_{k=2}^{n}\left(\frac{(k+2)(k-1)}{k(k+1)}\right)$$

$$\Rightarrow \frac{4.1}{2.3}\times\frac{5.2}{3.4}\times\frac{6.3}{4.5}\times.....\times\frac{(n+2)(n-1)}{n(n+1)}$$

$$\Rightarrow P_n \Rightarrow \frac{1}{3}\frac{(n+2)}{n}$$

$$\lim_{n\to\infty} P_n=\lim_{n\to\infty}\frac{1}{3}\frac{\left(1+\frac{2}{n}\right)}{1}=\frac{1}{3}=\frac{a}{b}$$

$$a+b=1+3=4$$

53) $f(x) = \{x\} + \{2x\} + \{3x\}$

$\{x\}$ is not differentiable at $x \in N$

$\Rightarrow x = \{1, 2, 3, ...\}$

$\{2x\}$ is not differentiable at $2x \in N$

$$\Rightarrow x=\left\{\frac{1}{2},\ 1,\ \frac{3}{2},\ 2,\ \frac{5}{2},\ ...\right\}$$

$\{3x\}$ is not differentiable at $3x \in N$

$$\Rightarrow x=\left\{\frac{1}{3},\ \frac{2}{3},\ 1,\ \frac{4}{3},\ \frac{5}{3},\ 2,\ ...\right\}$$

$\therefore f(x)$ is N.D. at

$$x\in\left\{\frac{1}{3},\ \frac{1}{2},\ \frac{2}{3},\ 1,\ \frac{4}{3},\ \frac{3}{2},\ \frac{5}{3},\ 2,\ ...\right\}$$

$\therefore f(x)$ is N.D. at 7 points in [0, 2)

$\therefore n = 2$ Ans.

54)
$$f(x)=\begin{cases} a_n+\cos\pi x & ,\ x\in\left[2n,\ 2n+\frac{2}{3}\right] \\ c_{n-2}+\sin\pi x+\cos\frac{\pi x}{2} & ,\ x\in\left(2n+\frac{2}{3},\ 2n+\frac{3}{2}\right] \\ b_{n+3}+\sin\frac{\pi x}{2} & ,\ x\in\left(2n+\frac{3}{2},\ 2n+2\right) \end{cases}$$

f(x) is continuous

$$a_n+\cos\frac{2\pi}{3}=c_{n-2}+\sin\frac{2\pi}{3}+\cos\left(n\pi+\frac{\pi}{3}\right) \quad ...(1)$$

and $$c_{n-2}+\sin\frac{3\pi}{2}+\cos\left(n\pi+\frac{3\pi}{4}\right)$$

$$=b_{n+3}+\sin\left(n\pi+\frac{3\pi}{4}\right) \quad ....(2)$$

and $b_{n+3} = a_{n+1} + \cos(n + 1)2\pi$ ....(3)

on add this equations

$$a_n - \frac{1}{2} - 1 + \cos\left(n\pi + \frac{3\pi}{4}\right)$$

$$= a_{n+1} + \frac{\sqrt{3}}{2} + \cos\left(n\pi + \frac{\pi}{3}\right) + \sin\left(n\pi + \frac{3\pi}{4}\right) + \cos(n+1)2\pi$$

if n odd $a_n - a_{n+1} = \frac{4 + \sqrt{3} - 2\sqrt{2}}{2}$

and if n even $a_n - a_{n+1} = \frac{6 + \sqrt{3} + \sqrt{8}}{2}$